§ 0.1

Kalman Filter

The Kalman filter is a special case of the Bayesian filter in which the state transition and observation models are linear and all uncertainties are Gaussian.

0.1.1 Prediction Step

Suppose that the state transition is linear in the previous state $\boldsymbol{x}_{t-1}$ and control input $\boldsymbol{u}_t$:

$$\boldsymbol{x}_t = \mathbf{F}_t \boldsymbol{x}_{t-1} + \mathbf{B}_t \boldsymbol{u}_t + \boldsymbol{q}_t, \qquad \boldsymbol{q}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_t), \qquad t \in \{1, \ldots, T\}.$$

Assume that the process noise $\boldsymbol{q}_t$ is independent of the previous state. Then

$$p(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}, \boldsymbol{u}_t) = \mathcal{N}\left(\mathbf{F}_t \boldsymbol{x}_{t-1} + \mathbf{B}_t \boldsymbol{u}_t, \mathbf{Q}_t\right).$$

Suppose that the posterior state distribution at time $t-1$ is Gaussian:

$$p(\boldsymbol{x}_{t-1} \mid \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t-1}) = \mathcal{N}\left(\boldsymbol{\mu}_{t-1|t-1}, \boldsymbol{\Sigma}_{t-1|t-1}\right),$$

where $\boldsymbol{\mu}_{t-1|t-1}$ and $\boldsymbol{\Sigma}_{t-1|t-1}$ denote the posterior mean and covariance based on all information available up to time $t-1$.

The prediction step of the Bayesian filter is

$$p(\boldsymbol{x}_t \mid \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) = \int p(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}, \boldsymbol{u}_t) p(\boldsymbol{x}_{t-1} \mid \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t-1}) \mathrm{d}\boldsymbol{x}_{t-1}.$$

Since $\boldsymbol{x}_{t-1}$ and the independent process noise $\boldsymbol{q}_t$ are Gaussian, and $\boldsymbol{x}_t$ is an affine transformation of them, the predictive distribution is also Gaussian:

$$p(\boldsymbol{x}_t \mid \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) = \mathcal{N}\left(\boldsymbol{\mu}_{t|t-1}, \boldsymbol{\Sigma}_{t|t-1}\right).$$

Its mean is

$$\begin{aligned}
\boldsymbol{\mu}_{t|t-1} &= \mathbb{E}[\boldsymbol{x}_t \mid \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}] && \leftarrow \text{Definition of predictive mean} \\
&= \mathbb{E}[\mathbf{F}_t \boldsymbol{x}_{t-1} + \mathbf{B}_t \boldsymbol{u}_t + \boldsymbol{q}_t] && \leftarrow \text{State transition model} \\
&= \mathbf{F}_t \mathbb{E}[\boldsymbol{x}_{t-1}] + \mathbf{B}_t \boldsymbol{u}_t + \mathbb{E}[\boldsymbol{q}_t] && \leftarrow \text{Linearity of expectation} \\
&= \mathbf{F}_t \boldsymbol{\mu}_{t-1|t-1} + \mathbf{B}_t \boldsymbol{u}_t && \leftarrow \text{Zero-mean process noise.}
\end{aligned}$$

Its covariance is

$$
\begin{aligned}
\boldsymbol{\Sigma}_{t|t-1} &= \mathrm{Cov}(\boldsymbol{x}_t \mid \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) && \leftarrow \text{Definition of predictive covariance} \\
&= \mathrm{Cov}(\mathbf{F}_t \boldsymbol{x}_{t-1} + \mathbf{B}_t \boldsymbol{u}_t + \boldsymbol{q}_t) && \leftarrow \text{State transition model} \\
&= \mathrm{Cov}(\mathbf{F}_t \boldsymbol{x}_{t-1} + \boldsymbol{q}_t) && \leftarrow \text{Deterministic control input} \\
&= \mathrm{Cov}(\mathbf{F}_t \boldsymbol{x}_{t-1}) + \mathrm{Cov}(\boldsymbol{q}_t) && \leftarrow \text{Independence of state and process noise} \\
&= \mathbf{F}_t \mathrm{Cov}(\boldsymbol{x}_{t-1}) \mathbf{F}_t^\top + \mathbf{Q}_t && \leftarrow \text{Covariance under linear transformation} \\
&= \mathbf{F}_t \boldsymbol{\Sigma}_{t-1|t-1} \mathbf{F}_t^\top + \mathbf{Q}_t.
\end{aligned}
$$

Therefore, the prediction step of the Kalman filter is

$$\boldsymbol{\mu}_{t|t-1} = \mathbf{F}_t \boldsymbol{\mu}_{t-1|t-1} + \mathbf{B}_t \boldsymbol{u}_t$$

and

$$\boldsymbol{\Sigma}_{t|t-1} = \mathbf{F}_t \boldsymbol{\Sigma}_{t-1|t-1} \mathbf{F}_t^\top + \mathbf{Q}_t.$$

0.1.2 Correction Step

Suppose that the observation model is also linear:

$$\boldsymbol{z}_t = \mathbf{H}_t \boldsymbol{x}_t + \boldsymbol{r}_t, \qquad \boldsymbol{r}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_t), \qquad t \in \{1, \ldots, T\},$$

where the observation noise $\boldsymbol{r}_t$ is independent of the state $\boldsymbol{x}_t$. Thus,

$$p(\boldsymbol{z}_t \mid \boldsymbol{x}_t, \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) = p(\boldsymbol{z}_t \mid \boldsymbol{x}_t) = \mathcal{N}\left(\mathbf{H}_t \boldsymbol{x}_t, \mathbf{R}_t\right).$$

We first show that the predicted state $\boldsymbol{x}_t$ and the observation $\boldsymbol{z}_t$ are jointly Gaussian. From the observation model,

$$\begin{bmatrix} \boldsymbol{x}_t \\ \boldsymbol{z}_t \end{bmatrix} = \begin{bmatrix} \boldsymbol{x}_t \\ \mathbf{H}_t \boldsymbol{x}_t + \boldsymbol{r}_t \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{H}_t & \mathbf{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{x}_t \\ \boldsymbol{r}_t \end{bmatrix}.$$

Before observing $\boldsymbol{z}_t$,

$$\boldsymbol{x}_t \sim \mathcal{N}\left(\boldsymbol{\mu}_{t|t-1}, \boldsymbol{\Sigma}_{t|t-1}\right),$$

and

$$\boldsymbol{r}_t \sim \mathcal{N}\left(\mathbf{0}, \mathbf{R}_t\right).$$

Since $\boldsymbol{x}_t$ and $\boldsymbol{r}_t$ are independent Gaussian random vectors,

$$\begin{bmatrix} \boldsymbol{x}_t \\ \boldsymbol{r}_t \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \boldsymbol{\mu}_{t|t-1} \\ \mathbf{0} \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{t|t-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_t \end{bmatrix} \right).$$

Because a linear transformation of a Gaussian random vector remains Gaussian, $\boldsymbol{x}_t$ and $\boldsymbol{z}_t$ are jointly Gaussian.

The predicted-state mean is

$$\boldsymbol{\mu}_{\boldsymbol{x}t} = \mathbb{E}[\boldsymbol{x}_t] = \boldsymbol{\mu}_{t|t-1}.$$

The predicted observation mean is

$$\boldsymbol{\mu}_{\boldsymbol{z}t} = \mathbb{E}[\boldsymbol{z}_t] = \mathbb{E}[\mathbf{H}_t \boldsymbol{x}_t + \boldsymbol{r}_t] = \mathbf{H}_t \mathbb{E}[\boldsymbol{x}_t] + \mathbb{E}[\boldsymbol{r}_t] = \mathbf{H}_t \boldsymbol{\mu}_{t|t-1}.$$

The predicted-state covariance is

$$\boldsymbol{\Sigma}_{\boldsymbol{x}\boldsymbol{x}t} = \mathrm{Cov}(\boldsymbol{x}_t) = \boldsymbol{\Sigma}_{t|t-1}.$$

The cross-covariance between the state and the observation is

$$\boldsymbol{\Sigma}_{\boldsymbol{x}\boldsymbol{z}t} = \mathrm{Cov}(\boldsymbol{x}_t, \boldsymbol{z}_t) = \mathrm{Cov}(\boldsymbol{x}_t, \mathbf{H}_t \boldsymbol{x}_t + \boldsymbol{r}_t) = \mathrm{Cov}(\boldsymbol{x}_t, \mathbf{H}_t \boldsymbol{x}_t) + \mathrm{Cov}(\boldsymbol{x}_t, \boldsymbol{r}_t) = \mathrm{Cov}(\boldsymbol{x}_t)\mathbf{H}_t^\top = \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top.$$

Similarly,

$$\boldsymbol{\Sigma}_{\boldsymbol{z}\boldsymbol{x}t} = \mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1}.$$

Finally, the observation covariance is

$$\boldsymbol{\Sigma}_{\boldsymbol{z}\boldsymbol{z}t} = \mathrm{Cov}(\boldsymbol{z}_t) = \mathrm{Cov}(\mathbf{H}_t \boldsymbol{x}_t + \boldsymbol{r}_t) = \mathbf{H}_t \mathrm{Cov}(\boldsymbol{x}_t)\mathbf{H}_t^\top + \mathrm{Cov}(\boldsymbol{r}_t) = \mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top + \mathbf{R}_t.$$

Therefore,

$$\begin{bmatrix} \boldsymbol{x}_t \\ \boldsymbol{z}_t \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \boldsymbol{\mu}_{t|t-1} \\ \mathbf{H}_t \boldsymbol{\mu}_{t|t-1} \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{t|t-1} & \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top \\ \mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} & \mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top + \mathbf{R}_t \end{bmatrix} \right).$$

The difference between the actual observation and its predicted mean is called the *innovation*:

$$\boldsymbol{\nu}_t = \boldsymbol{z}_t - \mathbb{E}[\boldsymbol{z}_t] = \boldsymbol{z}_t - \mathbf{H}_t \boldsymbol{\mu}_{t|t-1}.$$

Before observing $\boldsymbol{z}_t$, the innovation has zero mean:

$$\mathbb{E}[\boldsymbol{\nu}_t] = \mathbb{E}\left[\boldsymbol{z}_t - \mathbb{E}[\boldsymbol{z}_t]\right] = \mathbf{0}.$$

Its covariance is

$$\mathrm{Cov}(\boldsymbol{\nu}_t) = \mathrm{Cov}(\boldsymbol{z}_t) = \mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top + \mathbf{R}_t.$$

We define the *innovation covariance*

$$\mathbf{S}_t = \underbrace{\mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top}_{\text{uncertainty in the predicted observation}} + \underbrace{\mathbf{R}_t}_{\text{observation noise}}.$$

The two terms in $\mathbf{S}_t$ have distinct interpretations:

$$\underbrace{\mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top}_{\text{uncertainty in the predicted observation}} + \underbrace{\mathbf{R}_t}_{\text{observation noise}}.$$

Since $\boldsymbol{x}_t$ and $\boldsymbol{z}_t$ are jointly Gaussian, conditioning on the newly observed $\boldsymbol{z}_t$ gives a Gaussian posterior:

$$p(\boldsymbol{x}_t \mid \boldsymbol{z}_{1:t}, \boldsymbol{u}_{1:t}) = \mathcal{N}\left(\boldsymbol{\mu}_{t|t}, \boldsymbol{\Sigma}_{t|t}\right).$$

For jointly Gaussian random vectors, the conditional mean is

$$\begin{aligned}
\boldsymbol{\mu}_{t|t} &= \mathbb{E}[\boldsymbol{x}_t \mid \boldsymbol{z}_t] \\
&= \mathbb{E}[\boldsymbol{x}_t] + \mathrm{Cov}(\boldsymbol{x}_t, \boldsymbol{z}_t)\mathrm{Cov}(\boldsymbol{z}_t)^{-1}\left(\boldsymbol{z}_t - \mathbb{E}[\boldsymbol{z}_t]\right) && \leftarrow \text{Conditional Gaussian formula} \\
&= \boldsymbol{\mu}_{t|t-1} + \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top \mathbf{S}_t^{-1}\left(\boldsymbol{z}_t - \mathbf{H}_t \boldsymbol{\mu}_{t|t-1}\right) && \leftarrow \text{Substitution of mean and covariance.}
\end{aligned}$$

Define the *Kalman gain*

$$\mathbf{K}_t = \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top \mathbf{S}_t^{-1}.$$

The posterior mean can therefore be written compactly as

$$\boldsymbol{\mu}_{t|t} = \boldsymbol{\mu}_{t|t-1} + \mathbf{K}_t \boldsymbol{\nu}_t.$$

Thus, the new observation corrects the predicted mean by an amount proportional to the innovation $\boldsymbol{\nu}_t$, while the Kalman gain $\mathbf{K}_t$ determines how strongly the observation should influence the estimate.

For jointly Gaussian random vectors, the conditional covariance is

$$\begin{aligned}
\boldsymbol{\Sigma}_{t|t} &= \mathrm{Cov}(\boldsymbol{x}_t \mid \boldsymbol{z}_t) \\
&= \mathrm{Cov}(\boldsymbol{x}_t) - \mathrm{Cov}(\boldsymbol{x}_t, \boldsymbol{z}_t)\mathrm{Cov}(\boldsymbol{z}_t)^{-1}\mathrm{Cov}(\boldsymbol{z}_t, \boldsymbol{x}_t) && \leftarrow \text{Conditional Gaussian formula} \\
&= \boldsymbol{\Sigma}_{t|t-1} - \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top \mathbf{S}_t^{-1}\mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} && \leftarrow \text{Substitution of covariance terms} \\
&= \boldsymbol{\Sigma}_{t|t-1} - \mathbf{K}_t \mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1}.
\end{aligned}$$

Equivalently,

$$\mathbf{\Sigma}_{t|t} = (\mathbf{I} - \mathbf{K}_t \mathbf{H}_t)\, \mathbf{\Sigma}_{t|t-1}.$$

Therefore, the correction step of the Kalman filter can be summarized as

$$\begin{aligned} &\boldsymbol{\nu}_t = \boldsymbol{z}_t - \mathbf{H}_t \boldsymbol{\mu}_{t|t-1}, \quad \mathbf{S}_t = \mathbf{H}_t \mathbf{\Sigma}_{t|t-1} \mathbf{H}_t^\top + \mathbf{R}_t, \quad \mathbf{K}_t = \mathbf{\Sigma}_{t|t-1} \mathbf{H}_t^\top \mathbf{S}_t^{-1}, \\ &\boldsymbol{\mu}_{t|t} = \boldsymbol{\mu}_{t|t-1} + \mathbf{K}_t \boldsymbol{\nu}_t, \quad \mathbf{\Sigma}_{t|t} = (\mathbf{I} - \mathbf{K}_t \mathbf{H}_t)\, \mathbf{\Sigma}_{t|t-1}. \end{aligned}$$

Therefore, the new observation corrects the predicted state according to the discrepancy between the actual and predicted observations, while the magnitude of the correction depends on both the uncertainty of the state prediction and the observation noise. The complete Kalman filtering procedure is summarized in Algorithm 1.

Algorithm 1 Kalman Filter

Require: Initial posterior mean $\boldsymbol{\mu}_{0|0}$, initial posterior covariance $\mathbf{\Sigma}_{0|0}$, controls $\boldsymbol{u}_{1:T}$, observations $\boldsymbol{z}_{1:T}$, state transition matrices $\mathbf{F}_{1:T}$ and $\mathbf{B}_{1:T}$, observation matrices $\mathbf{H}_{1:T}$, process noise covariances $\mathbf{Q}_{1:T}$, and observation noise covariances $\mathbf{R}_{1:T}$

for $t = 1, 2, \ldots, T$ **do**
- $\boldsymbol{\mu}_{t|t-1} \leftarrow \mathbf{F}_t \boldsymbol{\mu}_{t-1|t-1} + \mathbf{B}_t \boldsymbol{u}_t$ ▷ Predicted mean
- $\mathbf{\Sigma}_{t|t-1} \leftarrow \mathbf{F}_t \mathbf{\Sigma}_{t-1|t-1} \mathbf{F}_t^\top + \mathbf{Q}_t$ ▷ Predicted covariance
- $\boldsymbol{\nu}_t \leftarrow \boldsymbol{z}_t - \mathbf{H}_t \boldsymbol{\mu}_{t|t-1}$ ▷ Innovation
- $\mathbf{S}_t \leftarrow \mathbf{H}_t \mathbf{\Sigma}_{t|t-1} \mathbf{H}_t^\top + \mathbf{R}_t$ ▷ Innovation covariance
- $\mathbf{K}_t \leftarrow \mathbf{\Sigma}_{t|t-1} \mathbf{H}_t^\top \mathbf{S}_t^{-1}$ ▷ Kalman gain
- $\boldsymbol{\mu}_{t|t} \leftarrow \boldsymbol{\mu}_{t|t-1} + \mathbf{K}_t \boldsymbol{\nu}_t$ ▷ Posterior mean
- $\mathbf{\Sigma}_{t|t} \leftarrow (\mathbf{I} - \mathbf{K}_t \mathbf{H}_t)\, \mathbf{\Sigma}_{t|t-1}$ ▷ Posterior covariance

end for

return $\left\{\boldsymbol{\mu}_{t|t}, \mathbf{\Sigma}_{t|t}\right\}_{t=1}^{T}$ of $\mathrm{bel}(\boldsymbol{x}_t)$